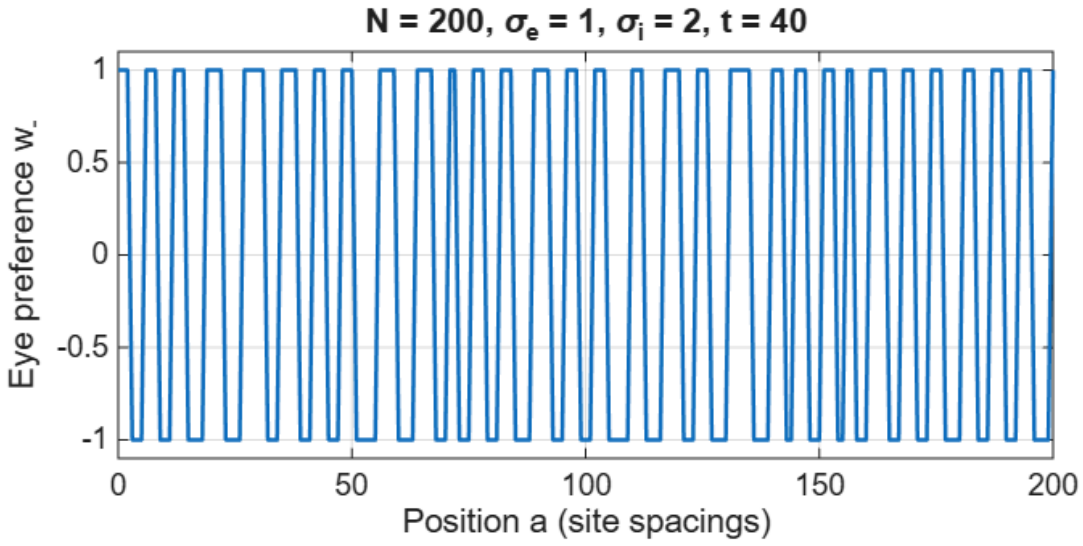
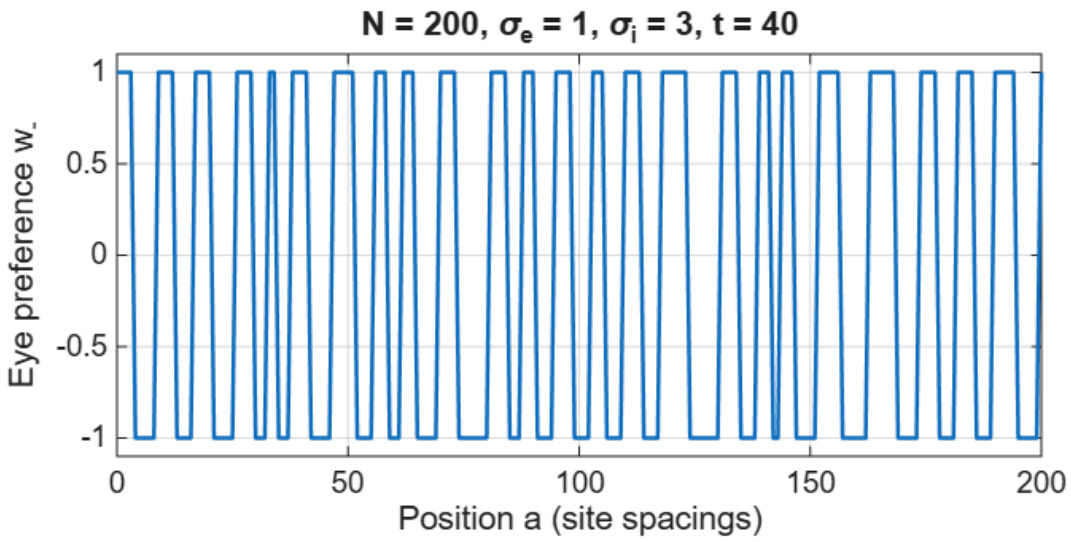
Computational Neuroscience homework has been uploaded to the following repo as per the instructions. Includes code as .m MATLAB files.

Account name: kywi4190
Repo url: https://github.com/kywi4190/CompNeuro2026
Email: kywi4190@colorado.edu

Figures
1c)



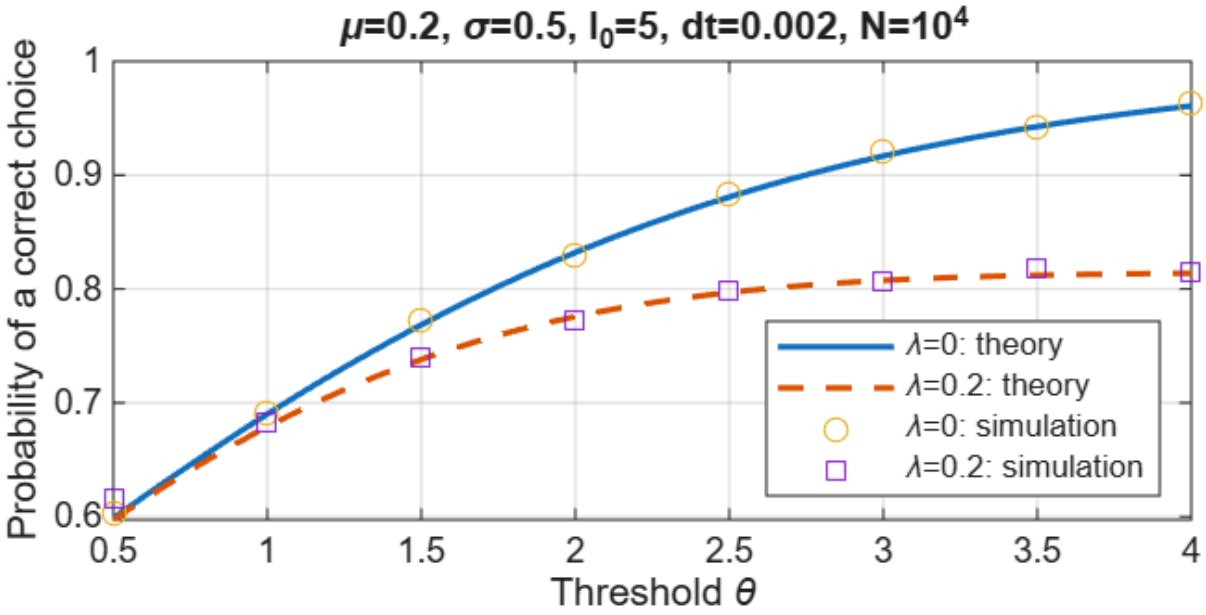


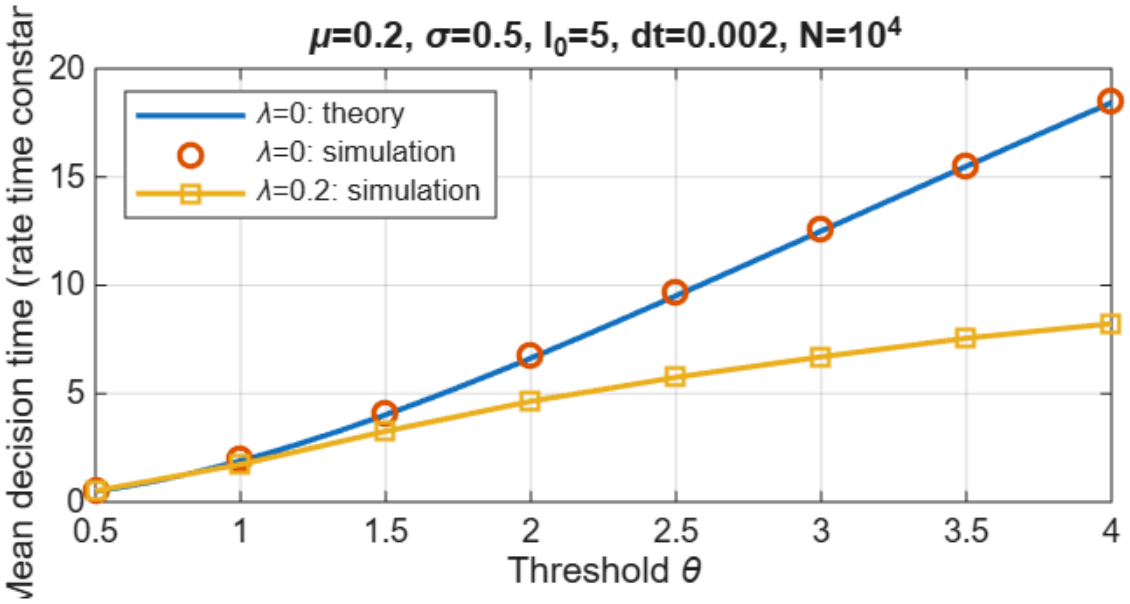
The stripe-pair periods, in site spacings, were:

sigma_i	Measured	Continuum Prediction	Fastest ring-mode prediction
2	6.456 +/- 0.3155	6.536	6.452
3	8.214 +/- 0.3795	8.478	8.333

The measured periods were close to both predictions, but closer to the fastest ring-mode prediction in both cases. Increasing sigma_i from 2 to 3 produced wider stripes.

2d)



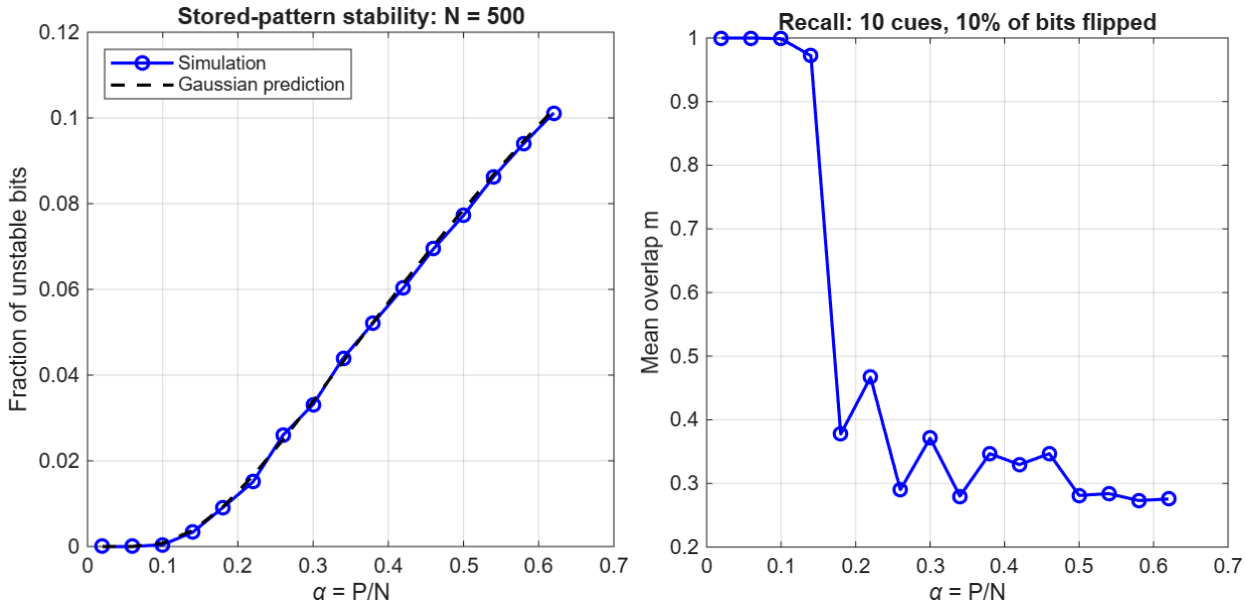


Unstable circuit: lambda=0.2, theta=4
Accuracy: 0.8138 (theory), 0.8155 (simulation)
Mean decision time: 8.23 (simulation)

Perfect integrator at the same predicted accuracy:
Threshold: 1.843
Accuracy: 0.8138
Mean decision time: 5.784 (theory)

For lambda=0.2 and theta=4 the simulated accuracy was 0.8155, close to the theoretical value 0.8138, with a mean decision time of 8.23. The perfect integrator achieves the same predicted accuracy at theta=1.843, with a predicted mean decision time of 5.784. Therefore the perfect integrator requires about 30% less time at the same predicted accuracy.

3c)



The fraction of unstable bits is very close to the theoretical prediction. Recall begins to break down around alpha=0.14-0.18, where the mean overlap drops noticeably below 1.

Homework 4

Thursday, October 15, 2026 6:33 PM

1. $v = w_R u_R + w_L u_L$

$\langle u_R^2 \rangle = \langle u_L^2 \rangle = q_s$

$\langle u_R u_L \rangle = q_D$

$\Big\} \; q_s > q_D > 0$

a) $\tau_w \frac{dw_R}{dt} = \langle v u_R \rangle$

$= \langle (w_R u_R + w_L u_L) u_R \rangle$

$= w_R \langle u_R^2 \rangle + w_L \langle u_L u_R \rangle$

$= q_s w_R + q_D w_L$

$\tau_w \frac{dw_L}{dt} = \langle v u_L \rangle$

$= w_R \langle u_R u_L \rangle + w_L \langle u_L^2 \rangle$

$= q_D w_R + q_s w_L$

with $w = (w_R, w_L)^T$

$\tau_w \frac{dw}{dt} = Qw \longrightarrow \boxed{Q = \begin{pmatrix} q_s & q_D \\ q_D & q_s \end{pmatrix}}$

$\det(Q - \lambda I) = \begin{vmatrix} q_s - \lambda & q_D \\ q_D & q_s - \lambda \end{vmatrix} = 0$

$(q_s - \lambda)^2 - q_D^2 = 0$

$(q_s - \lambda - q_D)(q_s - \lambda + q_D) = 0$

$\therefore \lambda_+ = q_s + q_d$

$\lambda_- = q_s - q_d$

eigenvectors:

$$(Q - \lambda_+ I)v = \begin{pmatrix} -q_d & q_d \\ q_d & -q_d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$-q_d x + q_d y = 0$$

$$y = x$$

$$\therefore v_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$(Q - \lambda_- I)v = \begin{pmatrix} q_d & q_d \\ q_d & q_d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$q_d x + q_d y = 0$$

$$y = -x$$

$$\therefore v_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

unconstrained:

$$w(t) = c_+(t) v_+ + c_-(t) v_-$$

$\rightarrow \tau_w w' = Qw$

$$\tau_w c_+' v_+ + \tau_w c_-' v_- = \lambda_+ c_+ v_+ + \lambda_- c_- v_-$$

$$\tau_w c_+' = \lambda_+ c_+ \qquad \tau_w c_-' = \lambda_- c_-$$

$$\rightarrow \tau_w c_\pm' = \lambda_\pm c_\pm \rightarrow c_\pm'(t) = \frac{\lambda_\pm}{\tau_w} c_\pm(t)$$

$$\frac{d}{dt}\left(e^{-\lambda_\pm t/\tau_w} c_\pm\right) = 0 \quad \rightarrow \text{integrating factor}$$

$$c_\pm(t) = c_\pm(0)\, e^{\lambda_\pm t/\tau_w}$$

$$w_R(0) = c_+(0) + c_-(0) \qquad w_L(0) = c_+(0) - c_-(0)$$

$$c_+(0) = \frac{w_R(0) + w_L(0)}{2} \qquad c_-(0) = \frac{w_R(0) - w_L(0)}{2}$$

$$w_R(t) = c_+(0)\, e^{(q_s+q_D)t/\tau_w} + c_-(0)\, e^{(q_s-q_D)t/\tau_w}$$

$$w_L(t) = c_+(0)\, e^{(q_s+q_D)t/\tau_w} - c_-(0)\, e^{(q_s-q_D)t/\tau_w}$$

$$\frac{w_R}{w_R}\left(\frac{\left[c_+(0)\, e^{(q_s+q_D)t/\tau_w}\right]^{-1}}{\left[c_+(0)\, e^{q_s+q_D t/\tau_w}\right]^{-1}}\right) = \frac{1 + \frac{c_-(0)}{c_+(0)}\, e^{-2q_D t/\tau_w}}{1 - \frac{c_-(0)}{c_+(0)}\, e^{-2q_D t/\tau_w}} \quad \rightarrow q_D > 0 \text{ so } e^{-2q_D t/\tau_w} \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$\therefore \lim_{t\to\infty} \frac{w_R(t)}{w_L(t)} = 1$$

Apply subtractive normalization:

$$\tau_w \frac{dw}{dt} = Qw - \frac{n \cdot Qw}{2} n \qquad n = (1,1)^T$$

$$n \cdot Qw = (q_s w_R + q_D w_L) + (q_D w_R + q_s w_L)$$

$$= (q_s + q_D)(w_R + w_L)$$

$$\tau_w \frac{dw_R}{dt} = q_s w_R + q_D w_L - \frac{q_s + q_D}{2}(w_R + w_L)$$

$$= \left(q_s - \frac{q_s+q_0}{2}\right) w_R + \left(q_0 - \frac{q_s+q_0}{2}\right) w_L$$

$$= \frac{q_s - q_0}{2}(w_R - w_L)$$

$$\tau_w \frac{dw_L}{dt} = q_0 w_R + q_s w_L - \frac{q_s+q_0}{2}(w_R + w_L)$$

$$= -\frac{q_s - q_0}{2} w_R + \frac{q_s - q_0}{2} w_L$$

$$= -\frac{q_s - q_0}{2}(w_R - w_L)$$

let $w_+ = w_R + w_L$

$w_- = w_R - w_L$

$$\tau_w \frac{dw_+}{dt} = 0$$

$$\therefore w_R(t) + w_L(t) = w_R(0) + w_L(0)$$

$$\tau_w \frac{dw_-}{dt} = \frac{q_s - q_0}{2} w_- - \left(-\frac{q_s - q_0}{2} w_-\right)$$

$$= (q_s - q_0) w_-$$

$$\therefore \tau_w w_-' = (q_s - q_0) w_-$$

$$w_-(t) = w_-(0)\, e^{(q_s - q_0)t/\tau_w}$$

let conserved sum $w_R + w_L = S$

we have $w_R - w_L = w_-(0)\, e^{(q_s - q_0)t/\tau_w}$

$$2w_R = S + w_-$$

$$w_R = \frac{s + w_-}{2}$$

$$2w_L = s - w_-$$

$$w_L = \frac{s - w_-}{2}$$

$$\therefore w_R = \frac{s + w_-(0)\, e^{(q_s - q_0)\, t/\tau_w}}{2}$$

$$w_L = \frac{s - w_-(0)\, e^{(q_s - q_0)\, t/\tau_w}}{2}$$

Impose Bounds:

$$0 \leq w_R, w_L \leq w_{max}$$

$$w_R + w_L = w_{max} > 0$$

$$w_R = \frac{w_{max} + w_-}{2}$$

$$w_L = \frac{w_{max} - w_-}{2}$$

Since $q_s > q_0 \rightarrow e^{(q_s - q_0)\, t/\tau_w}$ grows

So a non-zero w_- grows until it reaches a bound.

$$\therefore (w_R, w_L) \longrightarrow \begin{cases} (w_{max}, 0), & w_R(0) > w_L(0) \\ (0, w_{max}), & w_R(0) < w_L(0) \end{cases}$$

The one with the initial advantage wins.

The eye with the initial advantage wins.

Though if $w_R(0) = w_L(0) = \frac{w_{max}}{2}$ then the neuron remains binocular for all time.

Oja normalization:

$$\tau_w \frac{dw}{dt} = \langle vu - v^2 w \rangle = Qw - \langle v^2 \rangle w$$

$$v = w^T u \longrightarrow \langle v^2 \rangle = w^T (uu^T) w$$

$$= w^T Q w$$

$$\text{let } \beta(t) = w(t)^T Q w(t)$$

then:

$$\tau_w w_R' = (q_s - \beta) w_R + q_D w_L$$

$$\tau_w w_L' = q_D w_R + (q_s - \beta) w_L$$

$$\longrightarrow \tau_w w_+' = (\lambda_+ - \beta) w_+$$

$$\longrightarrow \tau_w w_-' = (\lambda_- - \beta) w_-$$

To determine eye preference:

$$r(t) = \frac{w_-(t)}{w_+(t)}$$

$$\tau_w r' = \frac{(\tau_w w_-') w_+ - w_- (\tau_w w_+')}{w_+^2}$$

$$= \frac{(\lambda_- - \beta) w_- w_+ - (\lambda_+ - \beta) w_- w_+}{w_+^2}$$

$$= (\lambda_- - \lambda_+) \frac{w_-}{w_+}$$

$= -2q_0 r$

So: $r(t) = r(0)\, e^{-2q_0 t/\tau_w}$

and: $\lim_{t\to\infty} r(t) = 0$

$$\frac{w_R}{w_L} = \frac{w_+ + w_-}{w_+ - w_-}\,\frac{\left(\frac{1}{2}\right)}{\left(\frac{1}{2}\right)} = \frac{1+\frac{w_-}{w_+}}{1-\frac{w_-}{w_+}} = \frac{1+r}{1-r}$$

$$\therefore \lim_{t\to\infty} \frac{w_R}{w_L} = 1$$

So Oja normalization causes neither eye to dominate and approaches binocular balance.

b) $v(a) = \sum_b K(a-b)\left[w_R(b)\, u_R + w_L(b)\, u_L\right]$

let $H_R(a) = \langle v(a)\, u_R \rangle$

$H_L(a) = \langle v(a)\, u_L \rangle$

$$H_R(a) = \sum_b K(a-b)\left[w_R(b)\langle u_R^2\rangle + w_L(b)\langle u_L u_R\rangle\right]$$

$$= \sum_b K(a-b)\left[q_s w_R(b) + q_0 w_L(b)\right]$$

Similarly: $H_L(a) = \sum_b K(a-b)\left[q_0 w_R(b) + q_s w_L(b)\right]$

apply subtractive normalization:

$$\tau_w \frac{\partial w_R(a)}{\partial t} = H_R(a) - \frac{H_R(a) + H_L(a)}{2}$$

$$\tau_w \frac{\partial w_L(a)}{\partial t} = H_L(a) - \frac{H_R(a) + H_L(a)}{2}$$

$$\tau_w \frac{\partial}{\partial t}[w_R(a) + w_L(a)] = H_R(a) + H_L(a) - [H_R(a) + H_L(a)] = 0$$

$\therefore\ w_R(a,t) + w_L(a,t) = w_R(a,0) + w_L(a,0)$ is conserved

$$\tau_w \frac{\partial w_-(a)}{\partial t} = H_R(a) - H_L(a)$$

$$= \sum_b k(a-b)[q_s w_R(b) + q_0 w_L(b) - q_0 w_R(b) - q_s w_L(b)]$$

$$= \sum_b k(a-b)[(q_s - q_0) w_R(b) - (q_s - q_0) w_L(b)]$$

$$\therefore \tau_w \frac{\partial w_-(a)}{\partial t} = (q_s - q_0) \sum_b k(a-b) w_-(b)$$

Growth rate:

$$\tau_w \frac{\partial w_-(a,t)}{\partial t} = (q_s - q_0) \int_{-\infty}^{\infty} k(a-b) w_-(b,t)\, db \quad \hookrightarrow \text{continuum limit}$$

and let $w_-(a,t) = A_k(t) e^{ika}$ $\hookrightarrow$ fourier ansatz

then $= (q_s - q_0) A_k(t) \int_{-\infty}^{\infty} k(a-b) e^{ikb}\, db$

let $x = a - b \rightarrow b = a - x$

$db = -dx$

for integral: $\int_{-\infty}^{\infty} k(a-b) e^{ikb}\, db = \int_{-\infty}^{\infty} k(x) e^{ik(a-x)}\, dx$

$$= e^{ika} \int_{-\infty}^{\infty} k(x)\, e^{-ikx}\, dx$$

$$= e^{ika}\, \tilde{k}(k) \quad \text{where} \quad \tilde{k}(k) = \int_{-\infty}^{\infty} k(x)\, e^{-ikx}\, dx$$

$$\tau_w A_k'(t)\, e^{ika} = (q_s - q_0) A_k(t)\, e^{ika}\, \tilde{k}(k)$$

$$A_k' = \frac{(q_s - q_0)\, \tilde{k}(k)\, A_k}{\tau_w}$$

$$\therefore A_k(t) = A_k(0)\, e^{\left[\frac{(q_s - q_0)}{\tau_w} \tilde{k}(k) t\right]}$$

grows at $\frac{(q_s - q_0)}{\tau_w} \tilde{k}(k)$

Mexican hat:

$$k(x) = e^{-x^2/(2\sigma_e^2)} - \frac{\sigma_e}{\sigma_i} e^{-x^2/(2\sigma_i^2)} \qquad \sigma_i > \sigma_e$$

use linearity to transform each Gaussian seperately:

let $G_\sigma(k) = \int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)}\, e^{-ikx}\, dx$

$G'_\sigma(k) = \int_{-\infty}^{\infty} (-ix)\, e^{-x^2/(2\sigma^2)}\, e^{-ikx}\, dx$ ⟶ valid because Gaussian and its product with $|x|$ are integrable

note: $\frac{d}{dx} e^{-x^2/(2\sigma^2)} = \frac{-x}{\sigma^2} e^{-x^2/(2\sigma^2)}$

⟶ $x e^{-x^2/(2\sigma^2)} = -\sigma^2 \frac{d}{dx} e^{-x^2/(2\sigma^2)}$

→ xc - dx

$$G'_\sigma(k) = i\sigma^2 \int_{-\infty}^{\infty} \frac{d}{dx}\left(e^{-x^2/(2\sigma^2)}\right) e^{-ikx}\, dx$$

$$\int_{-\infty}^{\infty} \frac{d}{dx}\left(e^{-x^2/(2\sigma^2)}\right) e^{-ikx}\, dx = \left[e^{-x^2/(2\sigma^2)} e^{ikx}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)} (-ik) e^{-ikx}\, dx$$

$$= ik\, G_\sigma(k)$$

$$G'_\sigma(k) = i\sigma^2 (ik\, G_\sigma(k))$$

$$= -\sigma^2 k\, G_\sigma(k)$$

$$\frac{d}{dk}\left[e^{\sigma^2 k^2/2} G_\sigma(k)\right] = 0$$

$$G_\sigma(k) = G_\sigma(0)\, e^{-\sigma^2 k^2/2}$$

To evaluate $G_\sigma(0)$ use:

$$y = \frac{x}{\sqrt{2}\sigma} \qquad dx = \sqrt{2}\sigma\, dy$$

$$\rightarrow \int_{-\infty}^{\infty} e^{-y^2}\, dy = \sqrt{\pi}$$

So: $G_\sigma(0) = \sqrt{2}\sigma \int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{2\pi}\,\sigma$

$$\therefore G_\sigma(k) = \sqrt{2\pi}\,\sigma\, e^{-\sigma^2 k^2/2}$$

by linearity: $\tilde{k}(k) = G_{\sigma_e}(k) - \frac{\sigma_e}{\sigma_i} G_{\sigma_i}(k)$

$$= \sqrt{2\pi}\,\sigma_e\, e^{-\sigma_e^2 k^2/2} - \frac{\sigma_e}{\sigma_i}\sqrt{2\pi}\,\sigma_i\, e^{-\sigma_i^2 k^2/2}$$

$$\therefore \tilde{k}(k) = \sqrt{2\pi}\,\sigma_e\left[e^{-\sigma_e^2 k^2/2} - e^{-\sigma_i^2 k^2/2}\right]$$

Note: $\tilde{k}(0) = 0$

and since $\sigma_i > \sigma_e$: $\tilde{k}(k) > 0$ for $k \neq 0$

Fastest growing wavenumber:

since $\frac{q_s - q_0}{\tau_w} > 0$

and $\sqrt{2\pi}\sigma_e > 0$

we equivalently maximize:

$$f(k) = e^{-\sigma_e^2 k^2/2} - e^{-\sigma_i^2 k^2/2}$$

$$f'(k) = -\sigma_e^2 k e^{-\sigma_e^2 k^2/2} + \sigma_i^2 k e^{-\sigma_i^2 k^2/2}$$

$$0 = k\left[\sigma_i^2 e^{-\sigma_i^2 k^2/2} - \sigma_e^2 e^{-\sigma_e^2 k^2/2}\right]$$

($k=0$ is a root

$$\sigma_i^2 e^{-\sigma_i^2 k^2/2} = \sigma_e^2 e^{-\sigma_e^2 k^2/2}$$

$$\frac{\sigma_i^2}{\sigma_e^2} e^{-(\sigma_i^2 - \sigma_e^2)k^2/2} = 1$$

$$\ln\left(\frac{\sigma_i^2}{\sigma_e^2}\right) - \frac{\sigma_i^2 - \sigma_e^2}{2} k^2 = 0$$

$$2\ln\left(\frac{\sigma_i}{\sigma_e}\right) = \frac{\sigma_i^2 - \sigma_e^2}{2} k^2$$

$$\therefore k_*^2 = \frac{4\ln(\sigma_i/\sigma_e)}{\sigma_i^2 - \sigma_e^2}$$

Check if it's a maximum:

$$= -\sigma_e^2 k^2/2 \left[\frac{\sigma_i^2}{} e^{-(\sigma_i^2 - \sigma_e^2)k^2/2} - 1\right]$$

$$F'(k) = k\sigma_e^2 e^{-\sigma_e^2 k^2/2}\left[\frac{\sigma_i^2}{\sigma_e^2} e^{-(\sigma_i^2-\sigma_e^2)k^2/2} - 1\right]$$

positive for $k > 0$

decreases strictly with k

So: $F'(k) > 0$ for $0 < k < K_*$

$F'(k) < 0$ for $k > K_*$

$\therefore K_*$ is positive global max

By evenness $-K_*$ has same growth rate

The fastest growing pattern is $W_R - W_L \propto \cos(K_* a)$

positive regions favor the right eye while negative regions favor the left. This gives alternating right-eye left-eye stripes.

For spacing a cosine repeats every 2π so one right-left pair has length $\frac{2\pi}{K_*}$. Each individual stripe is about $\frac{\pi}{K_*}$ wide.

The interaction widths σ_e and σ_i determine K_* so they set the spacing. Its not all one eye since a uniform preference corresponds to $k = 0$. Since $\tilde{k}(0) = 0$, that preference doesn't grow.

But $\tilde{k}(K_*) > 0$ so small alternating preferences grow into stripes instead.

2. $$\frac{dr_1}{dt} = -r_1 + w_s r_1 - w_c r_2 + I_0 + \frac{\mu}{2} + \sigma\xi_1(t)$$

$$\frac{dr_2}{dt} = -r_2 + w_s r_2 - w_c r_1 + I_0 - \frac{\mu}{2} + \sigma\xi_2(t)$$

$$\langle \xi_i(t)\,\xi_j(t')\rangle = \delta_{ij}\,\delta(t-t')$$

Fixed point and Eigenvalues

a) $\mu = \sigma = 0$

a) $\mu = \sigma = 0$

i) $0 = (w_s - 1) r_1^* - w_c r_2^* + I_0$

ii) $0 = (w_s - 1) r_2^* - w_c r_1^* + I_0$

$0 = (w_s - w_c - 1)(r_1^* + r_2^*) + 2 I_0$ → add eq

↳ $w_s = w_c$

$r_1^* + r_2^* = 2 I_0$

$0 = (w_s + w_c - 1)(r_1^* - r_2^*)$ → sub eq

let $\lambda = w_s + w_c - 1$

$\lambda (r_1^* - r_2^*) = 0$

∴ if $\lambda \neq 0$

$(r_1^*, r_2^*) = (I_0, I_0)$

if $\lambda = 0$

$r_1^* + r_2^* = 2 I_0$

line of fixed points

use Jacobian to determine if small perturbations cause growth or decay in fixed point

$$A = \begin{pmatrix} w_s - 1 & -w_c \\ -w_c & w_s - 1 \end{pmatrix}$$

$0 = \det(A - p I)$ → p eigen value

$$0 = \begin{vmatrix} w_s - 1 - p & -w_c \\ -w_c & w_s - 1 - p \end{vmatrix}$$

$$0 = (w_s - 1 - p)^2 - w_c^2$$

$$0 = (w_s - 1 - p - w_c)(w_s - 1 - p + w_c)$$

$p_s = w_s - w_c - 1$ $\quad p_d = w_s + w_c - 1$

→ $w_c = w_s$

$\therefore p_s = -1$

$p_d = \lambda$

$$A - p_s I = \begin{pmatrix} w_c & -w_c \\ -w_c & w_c \end{pmatrix}$$

$$\left(\begin{array}{cc|c} w_c & -w_c & 0 \\ -w_c & w_c & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$\therefore v_s = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$p_s = -1$

$$A - p_d I = \begin{pmatrix} -w_c & -w_c \\ -w_c & -w_c \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\therefore v_d = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$p_d = \lambda$

$s = r_1 + r_2$ and $d = r_1 - r_2$:

$$s' = (w_s - 1)(r_1 + r_2) - w_c(r_1 + r_2) + 2I_0 + \sigma(\xi_1 + \xi_2)$$

$$= (w_s - w_c - 1)s + 2I_0 + \sigma(\xi_1 + \xi_2)$$

$$= -s + 2I_0 + \sigma(\xi_1 + \xi_2)$$

$$d' = (w_s - 1)(r_1 - r_2) + w_c(r_1 - r_2) + \left(\frac{\mu}{2} + \frac{\mu}{2}\right) + \sigma(\xi_1 - \xi_2)$$

$$= (w_s + w_c - 1)d + \mu + \sigma(\xi_1 - \xi_2)$$

$$= \lambda d + \mu + \sigma(\xi_1 - \xi_2)$$

A unit white noise has zero mean and covariance $\delta(t - t')$

its integral over interval Δt has variance Δt

let $\xi_s(t) = \dfrac{\xi_1(t) + \xi_2(t)}{\sqrt{2}}$ ⟶ use independence of ξ_1 and ξ_2

and $\xi(t) = \dfrac{\xi_1(t) - \xi_2(t)}{\sqrt{2}}$

$$\langle \xi(t)\xi(t') \rangle = \frac{1}{2}\left[\langle \xi_1(t)\xi_1(t') \rangle - \langle \xi_1(t)\xi_2(t') \rangle - \langle \xi_2(t)\xi_1(t') \rangle + \langle \xi_2(t)\xi_2(t') \rangle\right]$$

$$= \frac{1}{2}\left[\delta(t - t') - 0 - 0 + \delta(t - t')\right]$$

$$= \delta(t - t')$$

similarly ξ_s has unit white noise intensity: $\langle \xi_s(t)\xi_s(t') \rangle = \delta(t - t')$

$$\langle \xi(t)\xi_s(t') \rangle = \frac{1}{2}\left[\delta(t - t') + 0 - \delta(t - t')\right] = 0$$

$\langle \xi_s(t)\xi(t') \rangle = \frac{1}{2}[\delta(t-t') - 0 + 0 - \delta(t-t')] = 0$

$$\therefore s' = -s + 2I_0 + \sqrt{2}\sigma\xi_s(t)$$

$$d' = \lambda d + \mu + \sqrt{2}\sigma\xi(t)$$

let $d' = -u'(d) + \text{noise}$

so $u'(d) = -\lambda d - \mu$

$$\therefore u(d) = \frac{-\lambda}{2}d^2 - \mu d + C$$

$\lambda \neq 0$:

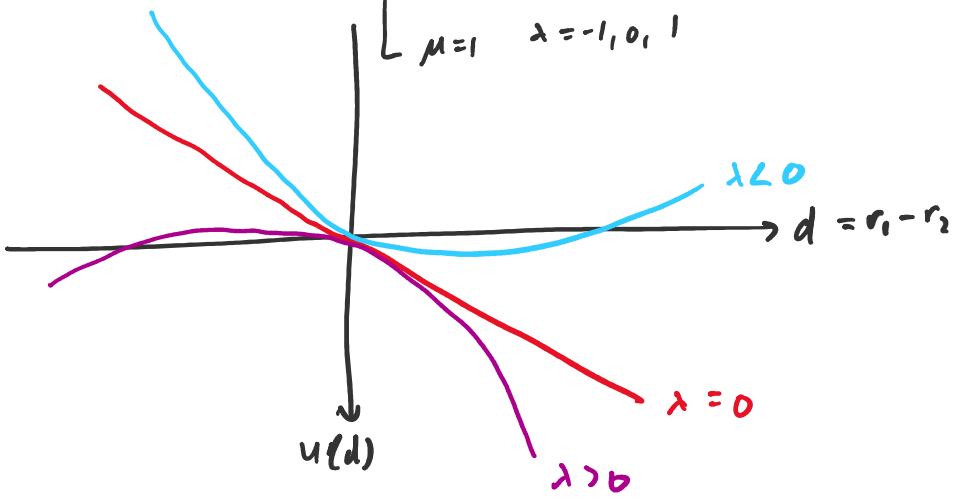
$$-\lambda d_* - \mu = 0$$

$$d_* = \frac{-\mu}{\lambda}$$

$$u'' = -\lambda$$

$u(d) = \frac{-\lambda}{2}d^2 - \mu d$

$\mu = 1 \quad \lambda = -1, 0, 1$



$$d' = \lambda d + \mu$$

$$d' - \lambda d = \mu$$

$$\frac{d}{dt}\left(e^{-\lambda t} d(t)\right) = \mu e^{-\lambda t} \quad \longrightarrow \text{integrate } 0 \to t$$

$$e^{-\lambda t} d t - d(0) = \mu \left[\frac{-e^{-\lambda u}}{\lambda} \right]_0^t$$

$$= \frac{\mu}{\lambda}\left(1 - e^{-\lambda t}\right)$$

$$\therefore d(t) = e^{\lambda t} d(0) + \frac{\mu}{\lambda}\left(e^{\lambda t} - 1\right)$$

$\lambda < 0$: potential is an upward parabola with a stable minimum at:

$$d_* = \frac{\mu}{|\lambda|} > 0$$

Starting from $d(0) = 0$:

$$d(t) = \frac{\mu}{|\lambda|}\left(1 - e^{-|\lambda| t}\right) \longrightarrow \frac{\mu}{|\lambda|}$$

Constant evidence causes $d(t)$ to approach a fixed value rather than grow indefinitely

$\lambda = 0$: $\qquad u(d) = -\mu d$

negative sloped line since $\mu > 0$

$d' = \mu \qquad d(t) = d(0) + \mu t$

The accumulated evidence neither grows or shrinks on its own. New evidence simply adds to the total.

$\lambda > 0$: downward parabola with unstable maximum at

$$d_* = \frac{-\mu}{\lambda} < 0$$

$\lambda > 0$: downward [illegible]

$$d_* = \frac{-\mu}{\lambda} < 0$$

$d < d_* \longrightarrow d' < 0$

$d > d_* \longrightarrow d' > 0$

Accumulated evidence is amplified, causing differences between populations to grow exponentially.

Drift diffusion:

$$\sigma^2 \pi'' + \mu \pi' = 0 \qquad \pi(-\theta) = 0 \qquad \pi(\theta) = 1$$

$$\sigma^2 T'' + \mu T' = -1 \qquad T(-\theta) = T(\theta) = 0$$

let $\rho(x) = \pi'(x)$

and $K = \frac{\mu}{\sigma^2} > 0$

$$\rho' + K\rho = 0$$

$$\frac{d}{dx}\left(e^{Kx}\rho(x)\right) = 0$$

$$\rho(x) = C e^{-Kx}$$

then $\pi(x) = A - \frac{C}{K} e^{-Kx}$

$$= A + B e^{-Kx}$$

$$\pi(\theta) = 1 = A + B e^{-K\theta}$$

$$\pi(-\theta) = 0 = A + B e^{K\theta}$$

$$B\left(e^{-K\theta} - e^{K\theta}\right) = 1$$

$$B = \frac{1}{e^{-k\theta} - e^{k\theta}}$$

$$A = -Be^{k\theta}$$

$$\pi(x) = \frac{e^{k\theta} - e^{-kx}}{e^{k\theta} - e^{-k\theta}}$$

$$\pi(x) = \frac{1 - e^{-\mu(x+\theta)/\sigma^2}}{1 - e^{-2\mu\theta/\sigma^2}}$$

For equal initial rates $x = 0$

let $a = \frac{\mu\theta}{\sigma^2}$

then $P(\text{correct}) = \pi(0)$

$$= \frac{1 - e^{-a}}{1 - e^{-2a}}$$

$$= \frac{1 - e^{-a}}{(1 - e^{-a})(1 + e^{-a})}$$

$$\therefore P(\text{correct}) = \frac{1}{1 + e^{-\mu\theta/\sigma^2}}$$

Solve for $T(x)$:

let $q(x) = T'(x)$

$$q' + kq = \frac{-1}{\sigma^2}$$

$$\frac{d}{\quad}(e^{kx} q(x)) = \frac{-1}{\sigma^2} e^{kx}$$

...

$$\frac{d}{dx}\left(e^{kx}q(x)\right) = \frac{-1}{\sigma^2}e^{kx}$$

$$e^{kx}q(x) = \frac{-e^{kx}}{\sigma^2 k} + c \longrightarrow \mu = \sigma^2 k$$

$$q(x) = \frac{-1}{\mu} + ce^{-kx}$$

$$T(x) = \frac{-x}{\mu} - \frac{c}{k}e^{-kx} + A$$

$$= \frac{-x}{\mu} + A + Be^{-kx}$$

$$0 = \frac{\theta}{\mu} + A + Be^{k\theta}$$

$$0 = \frac{-\theta}{\mu} + A + Be^{-k\theta}$$

$$0 = \frac{2\theta}{\mu} + B\left(e^{k\theta} - e^{-k\theta}\right)$$

$$B = \frac{-2\theta}{\mu\left(e^{k\theta} - e^{-k\theta}\right)}$$

$$A = \frac{-\theta}{\mu} - Be^{k\theta}$$

$$T(x) = -\frac{x+\theta}{\mu} + B\left(e^{-kx} - e^{k\theta}\right)$$

$$= -\frac{x+\theta}{\mu} + \frac{2\theta}{\mu}\frac{e^{k\theta} - e^{-kx}}{e^{k\theta} - e^{-k\theta}}$$

$$\therefore T(x) = \frac{2\theta\,\pi(x) - (x+\theta)}{\mu}$$

At $x=0$: $\langle T \rangle = T(0) = \frac{\theta}{\mu}\left(2\pi(0) - 1\right)$

$$= \frac{\theta}{\mu}\left(\frac{2}{1+e^{-a}} - 1\right)$$

$$= \frac{\theta}{\mu}\left(\frac{1-e^{-\Delta}}{1+e^{-\Delta}}\right)$$

$$= \frac{\theta}{\mu}\left(\frac{e^{\Delta/2}-e^{-\Delta/2}}{e^{\Delta/2}+e^{-\Delta/2}}\right) \rightarrow \text{by definition of tanh}$$

$$\therefore \langle T \rangle = \frac{\theta}{\mu}\tanh\left(\frac{\mu\theta}{2\sigma^2}\right)$$

If $\mu = 0$:

$$\sigma^2\pi'' = 0 \rightarrow \pi = Ax + B$$

$$-A\theta + B = 0 \qquad A\theta + B = 1$$

$$2A\theta = 1$$

$$A = \frac{1}{2\theta} \qquad B = \frac{1}{2}$$

$$\therefore \pi(x) = \frac{x+\theta}{2\theta} \qquad (\mu = 0)$$

$$T'' = \frac{-1}{\sigma^2}$$

$$T(x) = \frac{-x^2}{2\sigma^2} + c_1 x + c_2$$

$$0 = \frac{-\theta^2}{2\sigma^2} \pm c_1\theta + c_2$$

$$\rightarrow c_1 = 0 \qquad c_2 = \frac{\theta}{2\sigma^2}$$

$$\therefore T(x) = \frac{\theta^2 - x^2}{2\sigma^2} \qquad (\mu = 0)$$

$$\pi(0) = \frac{1}{2} \qquad \langle T \rangle = \frac{\theta^2}{2\sigma^2} \qquad (\mu = 0)$$

Speed Accuracy Tradeoff and Limits

For $\mu > 0$: $\dfrac{dP(\text{correct})}{d\theta} = \dfrac{\mu}{\sigma^2}\left(\dfrac{e^{-\mu\theta/\sigma^2}}{(1+e^{-\mu\theta/\sigma^2})^2}\right) > 0$

let $z = \mu\theta/2\sigma^2 > 0$

$$\frac{d\langle T \rangle}{d\theta} = \frac{1}{\mu}\left(\tanh z + z(1-\tanh^2 z)\right) > 0$$

So increasing θ increases both accuracy and mean-decision time. A larger threshold needs more accumulated evidence. This lowers the chance early noise determines the outcome but it takes longer.

Small thresholds: $\mu\theta/\sigma^2 \ll 1$

use $\tanh z = z + O(z^3)$

and $P(\text{correct}) = \frac{1}{2}\left[1 + \tanh\left(\frac{a}{2}\right)\right]$

then $P(\text{correct}) = \frac{1}{2}\left[1 + \frac{a}{2} + O(a^3)\right]$

$$= \frac{1}{2} + \frac{\mu\theta}{4\sigma^2} + O(\theta^3)$$

Similarly $\langle T \rangle = \frac{\theta}{\mu}\left[\frac{\mu\theta}{2\sigma^2} + O(\theta^3)\right]$

$$= \frac{\theta^2}{2\sigma^2} + O(\theta^4)$$

$\therefore$ as $\theta \to 0^+$: $P(\text{correct}) \to \frac{1}{2}$

$\langle T \rangle \to 0$

decisions are very fast but nearly random

large thresholds: $\mu\theta/\sigma^2 \gg 1$

$a \to \infty$ so $e^{-a} \to 0$

$$P(\text{correct}) = 1 - e^{-a} + O(e^{-2a})$$

$$\tan\left(\frac{\Delta}{2}\right) = \frac{1 - e^{-a}}{1 + e^{-a}} = 1 - 2e^{-a} + O(e^{-2a})$$

$\therefore$ as $\theta \to \infty$: $P(\text{correct}) \to 1$

$\langle T \rangle \to \infty$

error probability shrinks while mean decision time grows.

c) For $\lambda > 0$:

$$\sigma^2 \pi'' + (\lambda x + \mu)\pi' = 0 \qquad \pi(-b) = 0, \quad \pi(\theta) = 1$$

let $p(x) = \pi'(x)$

$$p' + \frac{\lambda x + \mu}{\sigma^2} p = 0$$

integrating factor: $M = c^{\int \frac{\lambda x + \mu}{\sigma^2} dx}$

$$= c^{\frac{\lambda x^2 + 2\mu x}{2\sigma^2}}$$

$$\frac{d}{dx}(Mp) = 0$$

$$= c^{-\left(\frac{\lambda x^2 + 2\mu x}{2\sigma^2}\right)}$$

$\frac{d}{dx}(\pi p) = 0$

$p = c\, e^{-\left(\frac{\lambda x^2 + 2\mu x}{2\sigma^2}\right)}$

let $g = e^{-\left(\frac{\lambda x^2 + 2\mu x}{2\sigma^2}\right)}$

$\pi' = cg$

$\pi(x) - \pi(-\theta) = c\int_{-\theta}^{x} g(y)\,dy$

$\pi(x) = c\int_{-\theta}^{x} g(y)\,dy$

$1 = c\int_{-\theta}^{\theta} g\,dy$

$$\therefore \pi(x) = \frac{\int_{0}^{x} g(y)\,dy}{\int_{-\theta}^{\theta} g(y)\,dy}$$

at $x=0$:

$$p(\text{correct}) = \frac{\int_{0}^{0} g\,dy}{\int_{-\theta}^{\theta} g\,dy}$$

$\lambda y^2 + 2\mu y = \lambda\left(y^2 + 2\frac{\mu}{\lambda}y\right)$

$= \lambda\left[\left(y + \frac{\mu}{\lambda}\right)^2 - \frac{\mu^2}{\lambda^2}\right]$

$= \lambda\left(y + \frac{\mu}{\lambda}\right)^2 - \frac{\mu^2}{\lambda}$

$\longrightarrow g(y) = \underbrace{e^{\frac{\mu^2}{2\lambda\sigma^2}}}_{\text{cancels}}\, e^{\left[\frac{-\lambda}{2\sigma^2}\left(y + \frac{\mu}{\lambda}\right)^2\right]}$

$$\longrightarrow p(\text{correct}) = \frac{\int_{-\theta}^{0} e^{\left[\frac{-\lambda}{2\sigma^2}\left(y + \frac{\mu}{\lambda}\right)^2\right]} dy}{\int_{-\theta}^{0} e^{\left[\frac{-\lambda}{2\sigma^2}\left(y + \frac{\mu}{\lambda}\right)^2\right]} dy}$$

let $z = \frac{\sqrt{\lambda}}{\sigma}(y + \frac{\mu}{\lambda})$

$dy = \frac{\sigma}{\sqrt{\lambda}} dz$

$y = -\theta \longrightarrow z_- = \frac{\mu - \lambda\theta}{\sigma\sqrt{\lambda}}$

$y = 0 \longrightarrow z_0 = \frac{\mu}{\sigma\sqrt{\lambda}}$

$y = \theta \longrightarrow z_+ = \frac{\mu + \lambda\theta}{\sigma\sqrt{\lambda}}$

then: $P(\text{correct}) = \dfrac{\int_{z_-}^{z_0} e^{-z^2/2} dz}{\int_{z_-}^{z_+} e^{-z^2/2} dz}$

→ standard normal CDF

$\Phi(a) = P(z \leq a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} e^{-z^2/2} dz \qquad z \sim N(0,1)$

$\int_a^b e^{-z^2/2} dz = \sqrt{2\pi}\,[\Phi(b) - \Phi(a)]$

$P(\text{correct}) = \dfrac{\Phi\left(\frac{\mu}{\sigma\sqrt{\lambda}}\right) - \Phi\left(\frac{\mu - \lambda\theta}{\sigma\sqrt{\lambda}}\right)}{\Phi\left(\frac{\mu + \lambda\theta}{\sigma\sqrt{\lambda}}\right) - \Phi\left(\frac{\mu - \lambda\theta}{\sigma\sqrt{\lambda}}\right)}$

as $\theta \to \infty$: $z_- \to -\infty \qquad z_+ \to +\infty$

z_0 fixed since $\Phi(-\infty) = 0 \qquad \Phi(+\infty) = 1$

$\therefore \lim_{\theta \to \infty} P(\text{correct}) = \Phi\left(\frac{\mu}{\sigma\sqrt{\lambda}}\right)$

Evidence

time-dependent eqn with noise retained

$$d\,d(t) = (\lambda\, d(t) + \mu)\,dt + \sqrt{2}\,\sigma\, dW(t)$$

$$d\left(e^{-\lambda t}\, d(t)\right) = \mu\, e^{-\lambda t}\, dt + \sqrt{2}\,\sigma\, e^{-\lambda t}\, dW(t)$$

$$\therefore\ d(t) = e^{\lambda t} d(0) + \int_0^t e^{\lambda(t-u)}\,\mu\, du + \sqrt{2}\,\sigma \int_0^t e^{\lambda(t-u)}\, dW(u)$$

an input arriving at time u is multiplied by $e^{\lambda(t-u)}$ by time t

For two equal increments arriving at $u_1 < u_2$ the ratio of effects at time t is:

$$\frac{e^{\lambda(t-u_1)}}{e^{\lambda(t-u_2)}} = e^{\lambda(u_2-u_1)} > 1$$

$\therefore$ for $\lambda > 0$ early evidence has the most influence on choice

Higher Threshold

We know $\lim_{\theta \to \infty} P(\text{correct}) = \Phi\left(\frac{\mu}{\sigma\sqrt{\lambda}}\right) < 1$

the limiting accuracy does not depend on θ so increasing an already large threshold has neglible improvement.

Homework 4.5

Wednesday, October 21, 2026 6:39 PM

3. the network state is:

$$s = (s_1, \cdots, s_N)$$

$$s_i \in \{-1, +1\}$$

local field at unit i:

$$h_i = \sum_{j=1}^{N} w_{ij} s_j$$

updates according to:

$$s_i \leftarrow sgn(h_i)$$

$$sgn(0) = 1$$

a) let $E(s) = \frac{-1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij} s_i s_j$

$$E(s) = \frac{-1}{2} \sum_{\substack{i \neq k \\ j \neq k}} w_{ij} s_i s_j$$ → separate terms

$$- \frac{1}{2} \sum_{j \neq k} w_{kj} s_k s_j$$

$$- \frac{1}{2} \sum_{i \neq k} w_{ik} s_i s_k$$

$$- \frac{1}{2} w_{kk} s_k^2$$

$w_{ii} = 0$ and $w_{ik} = w_{ki}$:

$w_{kk} = 0$ and $w_{ik} = w_{ki}$:

$$\sum_{i \neq k} w_{ik} s_i s_k = \sum_{j \neq k} w_{jk} s_j s_k = \sum_{j \neq k} w_{kj} s_k s_j$$

$$\longrightarrow E(s) = \frac{-1}{2} \sum_{\substack{i \neq k \\ j \neq k}} w_{ij} s_i s_j - s_k \sum_{j \neq k} w_{kj} s_j$$

let $E_{rest} = \frac{-1}{2} \sum_{\substack{i \neq k \\ j \neq k}} w_{ij} s_i s_j$

$$h_k = \sum_{j=1}^{N} w_{kj} s_j = \sum_{j \neq k} w_{kj} s_j \longrightarrow w_{kk} = 0$$

$$\longrightarrow E(s) = E_{rest} - s_k h_k$$

let s' be state after updating unit k:

$$s'_k = \text{sgn}(h_k) \qquad s'_j = s_j \quad (j \neq k)$$

Neither E_{rest} or h_k changes during this update

Zero-diagonal means:

$$E(s') = E_{rest} - s'_k h_k$$

$$\Delta E = E(s') - E(s)$$

$$= (E_{rest} - s'_k h_k) - (E_{rest} - s_k h_k)$$

$= -s'_k h_k + s_k h_k$

$\therefore \Delta E = -(s'_k - s_k) h_k$

update sets $s'_k = \text{sgn}(h_k)$ so:

$$s'_k h_k = \text{sgn}(h_k) h_k = |h_k|$$

$$\Delta E = s_k h_k - |h_k|$$

Since $s_k = \pm 1 \rightarrow s_k h_k = \pm |h_k|$

$\rightarrow s_k h_k \leq |h_k|$

$\therefore \Delta E \leq 0$

Suppose a flip occurs and $h_k \neq 0$:

$$s_k = -\text{sgn}(h_k)$$

$\rightarrow s_k h_k = -|h_k|$

$\rightarrow \Delta E = -|h_k| - |h_k|$

$\therefore \Delta E = -2|h_k| < 0$ for every flip with $h_k \neq 0$

For $h_k = 0 \rightarrow \Delta E = 0$

Asynchronous Updates

Lets assume every neuron keeps getting updated. Energy can only drop so many times.

It never increases and there are 2^N finitely many possible energy values.

After the final drop, only zero field flips remain. These change $-1 \to +1$ so each neuron can flip at most once more.

So flips eventually stop. Assuming ongoing updates the final state must be a fixed point.

Synchronous Updates

let $W = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Fields are $h_1 = -s_2 \quad h_2 = -s_1$

$s^{(0)} = (1,1) \to h_1^{(0)} = -1, \; h_2^{(0)} = -1$

$s^{(1)} = (-1,-1) \to h_1^{(1)} = 1, \; h_2^{(1)} = 1$

$s^{(2)} = (1,1) \to h_1^{(2)} = -1, \; h_2^{(2)} = -1$

$\therefore (1,1) \to (-1,-1) \to (1,1) \to \cdots$

$$E(s) = \frac{-1}{2}\left(w_{12} s_1 s_2 + w_{21} s_2 s_1\right)$$

$$= \frac{-1}{2}\left(-s_1 s_2 - s_2 s_1\right)$$

$$= s_1 s_2$$

$\longrightarrow E(1,1) = E(-1,-1) = 1$

constant energy

single-unit energy argument failure:

let $\delta_i = s_i' - s_i$

$s_i' = s_i + \delta_i$

then $\Delta E = \frac{-1}{2} \sum_{i \neq j} w_{ij}\left[(s_i + \delta_i)(s_j + \delta_j) - s_i s_j\right]$

$$= \frac{-1}{2} \sum_{i \neq j} w_{ij}\left(s_i \delta_j + \delta_i s_j + \delta_i \delta_j\right)$$

by symmetry:

$$\sum_{i \neq j} w_{ij} s_i \delta_j = \sum_{i \neq j} w_{ji} s_j \delta_i = \sum_{i \neq j} w_{ij} \delta_i s_j$$

$$\Delta E = -\sum_{i \neq j} w_{ij} \delta_i s_j - \frac{1}{2} \sum_{i \neq j} w_{ij} \delta_i \delta_j$$

$$= -\sum_i \delta_i h_i - \frac{1}{2} \sum_{i \neq j} w_{ij} \delta_i \delta_j$$

$$= -\sum_i \delta_i h_i - \frac{1}{2}\sum_{i\neq j} W_{ij}\,\delta_i\,\delta_j$$

$$\therefore \Delta E = -\sum_i \delta_i h_i - \frac{1}{2}\sum_{i\neq j} W_{ij}\,\delta_i\,\delta_j$$

For simultaneous updates cross terms with $i \neq j$ remain.

This is exactly what the single unit argument leaves out.

b) $W_{ij} = \frac{1}{N}\sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu \quad (i \neq j) \qquad W_{ii} = 0$

$$P(\xi_i^\mu = 1) = P(\xi_i^\mu = -1) = \frac{1}{2}$$

$E[\xi_i^\mu] = 0$ → expected value

$(\xi_i^\mu)^2 = 1$

Since $W_{ii} = 0$: $h_i = \sum_{j\neq i} W_{ij}\,\xi_j'$

$$h_i = \sum_{j\neq i}\left(\frac{1}{N}\sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu\right)\xi_j'$$

$$= \frac{1}{N}\sum_{j\neq i}\sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu \xi_j'$$

$$\xi_i' h_i = \frac{1}{N}\sum_{j\neq i}\sum_{\mu=1}^{p} \xi_i' \xi_i^\mu \xi_j^\mu \xi_j'$$

$$\xi_i' h_i = \frac{1}{N}\sum_{j\neq i} \xi_i' \xi_i' \xi_j' \xi_j' + \frac{1}{N}\sum_{j\neq i}\sum_{\mu=2}^{p} \xi_i' \xi_i^\mu \xi_j^\mu \xi_j'$$

$$\xi_i' \xi_i' \xi_j' \xi_j' = (\xi_i')^2 (\xi_j')^2 = 1$$

$N-1$ values of $j \neq i$:

$$\frac{1}{N} \sum_{j \neq i} 1 = \frac{N-1}{N}$$

let $C_i = \frac{1}{N} \sum_{j \neq i} \sum_{\mu=2}^{p} \xi_i^1 \xi_i^\mu \xi_j^\mu \xi_j^1$

$$\therefore \xi_i^1 h_i = \frac{N-1}{N} + C_i$$

for fixed unit i let: $Y_{j\mu} = \xi_i^1 \xi_i^\mu \xi_j^\mu \xi_j^1 \quad (j \neq i) \quad \mu \geq 2$

$$\longrightarrow C_i = \frac{1}{N} \sum_{j \neq i} \sum_{\mu=2}^{p} Y_{j\mu}$$

$$\mathbb{E}[Y_{j\mu}] = \mathbb{E}[\xi_i^1]\,\mathbb{E}[\xi_i^\mu]\,\mathbb{E}[\xi_j^\mu]\,\mathbb{E}[\xi_j^1] = 0$$

$$\longrightarrow \mathbb{E}[C_i] = \frac{1}{N} \sum_{j \neq i} \sum_{\mu=2}^{p} \mathbb{E}[Y_{j\mu}] = 0$$

$$\therefore \mathbb{E}[C_i] = 0$$

Since $\mathbb{E}[C_i] = 0$:

$$\mathrm{Var}(C_i) = \mathbb{E}[C_i^2]$$

$$= \frac{1}{N^2} \sum_{j \neq i} \sum_{k \neq i} \sum_{\mu=2}^{p} \sum_{\nu=2}^{p} \mathbb{E}[Y_{j\mu} Y_{k\nu}]$$

$$Y_{j\mu} Y_{k\nu} = (\xi_i^1 \xi_i^\mu \xi_j^\mu \xi_j^1)(\xi_i^1 \xi_i^\nu \xi_k^\nu \xi_k^1)$$

$$= \xi_j^\mu \xi_i^\nu \xi_i^\mu \xi_k^\nu \xi_j^1 \xi_k^1 \longrightarrow \text{where } (\xi_i^1)^2 = 1$$

$$= \xi_i^\mu \xi_i^\nu \xi_j^\mu \xi_k^\nu \xi_j' \xi_k' \quad \longrightarrow \text{where } (\xi_i')^2 = 1$$

case $\mu \neq \nu$: $\mathbb{E}[Y_{j\mu} Y_{k\nu}] = \mathbb{E}[\xi_j' \xi_k'] \, \mathbb{E}[\xi_i^\mu \xi_j^\mu] \, \mathbb{E}[\xi_i^\nu \xi_k^\nu]$ $\quad (j \neq i)$

$$\mathbb{E}[\xi_i^\mu \xi_j^\mu] = \mathbb{E}[\xi_i^\mu] \mathbb{E}[\xi_j^\mu] = 0$$

$$\longrightarrow E[Y_{j\mu} Y_{k\nu}] = 0$$

case $\mu = \nu$, $j \neq k$: $Y_{j\mu} Y_{k\mu} = (\xi_i^\mu)^2 \xi_j^\mu \xi_k^\mu \xi_j' \xi_k'$

$$= \xi_j^\mu \xi_k^\mu \xi_j' \xi_k'$$

$$\longrightarrow \mathbb{E}[Y_{j\mu} Y_{k\mu}] = 0$$

case $\mu = \nu$, $j = k$: $Y_{j\mu}^2 = (\xi_i')^2 (\xi_i^\mu)^2 (\xi_j^\mu)^2 (\xi_j')^2 = 1$

$$\longrightarrow \mathbb{E}[Y_{j\mu}^2] = 1$$

only terms with matching pairs survive

$N-1$ choices of j

$P-1$ choices of μ

$$\text{var}(C_i) = \frac{1}{N^2} \sum_{j \neq i}^{N} \sum_{\mu=2}^{P} 1$$

$$= \frac{(N-1)(P-1)}{N^2}$$

$$\therefore \mathbb{E}[C_i] = 0 \qquad Var(C_i) = \frac{(N-1)(P-1)}{N^2}$$

using $P = \alpha N$:

$$Var(C_i) = \left(1 - \frac{1}{N}\right)\left(\frac{P}{N} - \frac{1}{N}\right)$$

$$= \left(1 - \frac{1}{N}\right)\left(\alpha - \frac{1}{N}\right)$$

For large N $\quad Var(C_i) \approx \alpha$

instability Probability:

The crosstalk is a sum of: $M = (N-1)(P-1)$

random signs divided by N.

For large M CLT gives:

$$\frac{\sum_{j \neq i} \sum_{\mu=2}^{P} y_{j\mu}}{\sqrt{M}} \approx z \qquad z \sim N(0,1)$$

Since $C_i = \frac{1}{N} \sum_{j \neq i} \sum_{\mu=2}^{P} y_{j\mu}$

$$\longrightarrow C_i \approx \mathcal{N}\left(0, \frac{(N-1)(P-1)}{N^2}\right)$$

with large N $\quad C_i \approx \mathcal{N}(0, K)$

let the margin $\quad m_i = \xi_i^1 h_i$

$m_i < 0 \longrightarrow$ negative margin means local field opposes stored bit

$$m_i < 0 \longleftrightarrow C_i < -\frac{N-1}{N}$$

let $\quad \sigma_C = \frac{\sqrt{(N-1)(P-1)}}{N}$

$$\text{Prob}(m_i < 0) = \text{Prob}\left(C_i < -\frac{N-1}{N}\right)$$

$$\approx \text{Prob}\left(z < -\frac{(N-1)}{N}\left(\frac{N}{\sqrt{(N-1)(P-1)}}\right)\right)$$

$$= \text{Prob}\left(z < \sqrt{\frac{N-1}{P-1}}\right)$$

$$q_N = \text{Prob}(\text{unstable bit}) \approx \Phi\left(-\sqrt{\frac{N-1}{P-1}}\right)$$

for large N, P with $P = \alpha N$

$$\sqrt{\frac{N-1}{P-1}} \approx \sqrt{\frac{N}{P}} = \frac{1}{\sqrt{\alpha}}$$

$$\therefore q = \text{Prob}(\text{unstable bit}) \approx \Phi\left(\frac{-1}{\sqrt{\alpha}}\right)$$

instability prob = 1%

$\bar{D}(z)$

instability prob = 1%

$\Phi\left(\frac{-1}{\sqrt{\alpha}}\right) = 0.01$ → $\Phi(-z) = 1 - \Phi(z)$

$\Phi\left(\frac{1}{\sqrt{\alpha}}\right) = 0.99$

$\Phi^{-1}(0.99) \approx 2.33$

$\frac{1}{\sqrt{\alpha}} \approx 2.33$

$\therefore \alpha \approx 0.1848$

$p \approx 0.1848N$

at 1%

every bit stable:

let A_i be event that bit i unstable

let U be number unstable bits

$\mathbb{E}[U] = \sum_{i=1}^{N} \text{Prob}(A_i) = Nq$ → marginal instability probability q

$\text{prob}\left(\bigcup_{i=1}^{N} A_i\right) \leq \sum_{i=1}^{N} \text{Prob}(A_i) = Nq$

$\therefore$ Prob (at least one unstable bit) $\leq Nq$

Prob (every bit stable) $\geq 1 - Nq$

α be stable

Prob (every

for entire pattern to be stable

$Nq \to 0$

for small q we need small α

So $z = \frac{1}{\sqrt{\alpha}}$ is large

$$\Phi(-z) \approx \frac{e^{-z^2/2}}{z\sqrt{2\pi}}$$

$$z^2 = \frac{1}{\alpha} \qquad \sqrt{\alpha} = \frac{1}{z}$$

$$q \approx \frac{\sqrt{\alpha}}{\sqrt{2\pi}} e^{-1/2\alpha}$$

$$\to Nq \approx \frac{N\sqrt{\alpha}}{\sqrt{2\pi}} e^{-1/2\alpha} \to P = \alpha N$$

$$Nq \approx \frac{N}{\sqrt{2\pi}} \sqrt{\frac{P}{N}}\, e^{-N/2P}$$

$$= \sqrt{\frac{NP}{2\pi}}\, e^{-N/2P}$$

Set $Nq \approx \varepsilon \qquad 0 < \varepsilon \ll 1$

$$N \frac{e^{-z^2/2}}{z\sqrt{2\pi}} \approx \varepsilon$$

$$\ln N - \frac{z^2}{2} - \ln z - \frac{1}{2}\ln(2\pi) \approx \ln \varepsilon$$

$$\frac{z^2}{2} + \ln z \approx \ln N - \ln \varepsilon - \frac{1}{2}\ln(2\pi)$$

$$\longrightarrow \frac{z^2}{2} + \ln z = \frac{z^2}{2}\left(1 + \frac{2\ln z}{z^2}\right) \sim \frac{z^2}{2}$$

$$\longrightarrow \frac{z^2}{2} \sim \ln N$$

$$\frac{N}{2p} \sim \ln N$$

$$\therefore \; p \sim \frac{N}{2\ln N}$$

$$\alpha \sim \frac{1}{2\ln N}$$